Echo State Networks in Reservoir Computing: Foundations, Benchmarks, and Applications to Next-G Wireless Communication

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***Abstract*— Echo State Networks (ESNs) are a lightweight reservoir computing model well-suited for time-series datasets, due to their sparsely connected reservoir layer and linear readout layer. This paper bridges ESN theory, benchmarking, and application to wireless communications for channel prediction. We first review the theory and mathematical formulation of ESNs. We then evaluate their performance on a public time-series benchmark — Mackey-Glass. The results show that ESNs can achieve low normalized mean squared error with a significantly reduced number of training parameters. Building on these insights, we design an ESN-based framework for wireless channel prediction that learns the dynamics of rapidly changing channel conditions and predicts the current channel based on outdated observations. Simulation results demonstrate that the ESN-based approach outperforms a traditional auto-regression method in prediction accuracy. Taken as a whole, these analyses highlight ESNs as a principled, efficient, and deployable solution for temporal inference and channel prediction in modern wireless systems, which could be very helpful for 5G/6G communications.**

Keywords—Echo State Network; Reservoir Computing; Temporal Data Processing, Wireless Communication

I. Introduction

Time-series data prediction plays a critical role in numerous real-world applications, ranging from financial forecasting and environmental monitoring to modern wireless communication systems. These scenarios demand models capable of learning complex temporal dependencies while remaining efficient enough for real-time deployment. As wireless systems evolve toward higher mobility and lower latency, the need for accurate and computationally lightweight prediction models becomes increasingly pronounced. In the context of wireless communications, accurate channel prediction is particularly important for 5G and beyond-5G (6G) networks. Reliable prediction of these variations enables proactive resource allocation, beamforming, and handover decisions, thereby improving spectral efficiency, reducing latency, and ensuring quality of service. As wireless systems continue to evolve, the demand for accurate yet computationally lightweight prediction models will only become more critical for the practical realization of next-generation communication systems.

Traditional statistical methods, such as auto-regression (AR), provide simple yet interpretable models for time-series prediction [1]. However, their linear structures often limit their ability to capture the nonlinear dynamics present in rapidly changing environments. Deep learning models like recurrent neural networks (RNNs) and long short-term memory networks (LSTMs) address this issue but introduce challenges such as heavy computational costs, large training datasets, and long convergence times [2].

Echo State Networks (ESNs), a key variant of reservoir computing, offer a compelling balance between model simplicity and expressive power [3]. ESNs feature a fixed, sparsely connected reservoir layer that transforms input signals into a high-dimensional space, followed by a linear readout layer trained using efficient regression methods [4]. This architecture drastically reduces the number of trainable parameters while retaining the ability to model complex temporal dependencies.

This work integrates three aspects: (i) a clear exposition of ESN theory and its mathematical formulation, (ii) rigorous benchmarking on a widely used time-series dataset -- Mackey-Glass, and (iii) a practical application to wireless channel prediction, where real-time adaptability is crucial. Together, these steps establish both the theoretical foundation and empirical benefits of ESNs for temporal inference tasks.

II. Background

A. Reservoir Computing

Reservoir Computing (RC) is a computational framework that emerged as an effective method to harness the power of recurrent neural networks (RNNs) while bypassing their traditionally difficult and computationally expensive training procedures. The core principle of RC is to decouple the recurrent part of the network from the trainable output layer.

RC was developed in the early 2000s as a novel paradigm to circumvent the challenges associated with training conventional RNNs, such as the vanishing and exploding gradients problem and high computational costs [3]. The key idea is that a randomly generated, fixed recurrent network—the reservoir—can act as a rich temporal feature extractor. By projecting an input signal into its high-dimensional state space,

2025 IEEE International Conference on Data Mining Workshops (ICDMW) | 979-8-3315-8132-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDMW69685.2025.00405

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the reservoir transforms the temporal patterns in the data into a format that is often linearly separable. This allows the complex, nonlinear processing to be handled by the fixed reservoir, while learning is simplified to training a linear readout layer. This separation leads to significant advantages, including extremely fast, non-iterative training, reduced data requirements, and a lower risk of poor local minimum, making RC particularly well-suited for real-time applications [5] and hardware implementation [6, 7, 8].

Crucially, the internal connections within the reservoir are not trained; they are initialized randomly and remain fixed. The only part of the system that undergoes training is the readout layer, which is typically optimized using a simple and efficient method like linear regression. This approach offers significant advantages over traditional RNNs, including dramatically faster training times, lower computational complexity, and a natural suitability for low-power hardware implementations. The two primary models that dominate the RC field are Echo State Networks (ESNs), which use rate-based neurons, and Liquid State Machines (LSMs) [9], which are based on spiking neurons.

III. Echo State Networks (ESNs)

ESNs are a specific type of RNNs within the reservoir computing paradigm, notable for their training efficiency and ability to model complex temporal dynamics. The core of an ESN is a large, fixed, randomly generated recurrent neural network that nonlinearly projects an input signal into a higher-dimensional computational space.

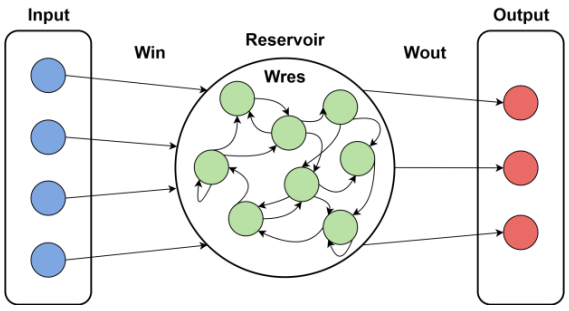


Fig. 1. The general architecture of an Echo State Network

The architecture of an ESN, as illustrated in Fig. 1, consists of three main components: an input layer, the recurrent "reservoir," and a linear "readout" layer. The connections from the input to the reservoir are defined by the input weight matrix W_{in}, and the internal recurrent connections within the reservoir are defined by the reservoir weight matrix W_{res}. Both W_{in} and W_{res} are randomly initialized and remain fixed. The only component that is modified during the training process is the output weight matrix W_{out}, which connects the reservoir states to the output layer. This separation of a fixed, nonlinear dynamic system from a simple, trainable linear output is the key to the ESN's remarkable efficiency.

A. Mathematical Foundations

The dynamics of the ESN's reservoir are governed by a discrete-time state update equation:

$$x(t+1) = f\big(W_{in}\, u(t+1) + W_{rec}\, x(t)\big) \quad (1)$$

Where:

- $x(t) \in \mathbb{R}^N$ is the vector of reservoir neuron activations at time *t*, with *N* being the number of neurons in the reservoir.
- $u(t) \in \mathbb{R}^M$ is the input vector at time *t*.
- $W_{in} \in \mathbb{R}^{N\times M}$ is the input weight matrix, which connects the input layer to the reservoir.
- $W_{rec} \in \mathbb{R}^{N\times N}$ is the internal reservoir weight matrix, defining the recurrent connections within the reservoir.
- f is the element-wise non-linear activation function, which is typically the hyperbolic tangent (tanh) function.

A fundamental requirement for a functional ESN is the echo state property (ESP). This property ensures that the effect of any initial state of the reservoir, $x(0)$, fades over time, making the reservoir's state $x(t)$ uniquely dependent on the history of the input signal u. A common method to ensure the ESP is to scale the reservoir matrix W_{rec} such that its spectral radius $\rho(W_{rec})$—the largest absolute eigenvalue—is less than 1 [3], [4]. This keeps the reservoir dynamics stable.

The network's output is generated by a linear readout layer that takes the reservoir's state as input:

$$y(t) = W_{out}[x(t); u(t)] \quad (2)$$

Here, $y(t) \in \mathbb{R}^L$ is the output vector and $W_{out} \in \mathbb{R}^{L\times(N+M)}$ is the output weight matrix. The notation $[x(t); u(t)]$ indicates the concatenation of the reservoir state and the input vector. Crucially, W_{in} and W_{rec} are initialized randomly and remain fixed throughout the process. Only the readout matrix W_{out} is trained.

B. Key Hyperparameters

The performance of an ESN is highly dependent on a set of hyperparameters that define the structure and dynamics of the reservoir. As outlined in the abstract, our work benchmarks the following key parameters:

- **Reservoir Size**: The number of neurons in the reservoir. A larger reservoir can capture more complex and longer-term dependencies in the input data but increases computational cost.
- **Spectral Radius**: Controls the memory capacity of the ESN. A value close to 1.0 allows for a longer memory of past inputs, which is critical for time-series prediction tasks.
- **Sparsity**: The proportion of zero-valued connections in the W_{rec} matrix. A sparse reservoir can foster more diverse and independent internal dynamics.
- **Input Scaling**: A scaling factor applied to the input signal. This parameter controls the degree to which the input drives the reservoir dynamics and influences the extent of non-linearity in the neuron activations.

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- **Learning Rate**: Although training is typically done via linear regression, iterative learning methods can also be used. In such cases, the learning rate determines the step size for updating the readout weights (W_{out}).

C. Training the Readout Layer

The primary advantage of ESNs is the simplicity and efficiency of their training process. Since the reservoir is fixed, training is reduced to finding the optimal weights for the linear readout layer, W_{out}. This is achieved through a straightforward supervised learning method, typically linear regression.

The training process involves two stages:

1. **State Collection:** The training data is fed into the reservoir, and the corresponding internal states $x(t)$ are collected for all time steps into a state matrix X .
2. **Linear Regression:** The output weights W_{out} are computed by solving a linear regression problem to minimize the error between the network output Y and the target output Y_{target}. This is often done using ridge regression (Tikhonov regularization) to prevent overfitting [11]:

$$W_{out} = Y_{t\arg et} X^T (XX^T + \beta I)^{-1} \qquad (3)$$

where β is a regularization coefficient and I is the identity matrix. This closed-form solution avoids the computationally expensive and gradient-related issues (e.g., vanishing / exploding gradients) of backpropagation through time used in conventional RNNs.

IV. Performance Evaluation on Benchmarks

To systematically evaluate the performance of our ESN framework and quantify the trade-offs associated with its hyperparameters, we use two canonical benchmarks in nonlinear time-series processing.

A. Mackey-Glass

The Mackey-Glass system is a time-delayed nonlinear differential equation known for exhibiting chaotic behavior [10]. It is a standard benchmark for testing the predictive capabilities of neural networks on chaotic time-series data [12]. The system is defined as follows:

$$y'(t) = \frac{\alpha y(t-\tau)}{1+(y(t-\tau))^{\beta}} - \gamma y(t) \qquad (4)$$

where $\alpha = 0.2$, $\beta = 10$, $\gamma = 0.1$, and the time delay τ is 17.

We performed a systematic evaluation by training and testing our ESN model with various reservoir sizes, ranging from 10 to 200 neurons, to identify the optimal configuration for this task. Our analysis of the hyperparameter sweep revealed that increasing the number of reservoir neurons does not necessarily lead to better prediction accuracy on unseen data. While the training error consistently decreased with larger reservoirs, the generalization performance on the test set was optimal with a relatively small reservoir of 20 neurons. This suggests that larger models begin to overfit the chaotic dynamics present in the training data. This result highlights a key advantage of the ESN architecture: its ability to capture complex temporal dynamics effectively with a computationally efficient model, avoiding the need for excessively large networks.

We conducted a series of experiments to evaluate the ESN's performance on the Mackey-Glass prediction task, focusing on the impact of the reservoir size. Table I shows the training and testing Normalized Mean Squared Error (NMSE) for ESNs with different numbers of reservoir neurons.

TABLE I. NMSE Performance vs. Number of Reservoir Neurons

	Training MSE	Training NMSE	Testing MSE	Testing NMSE
N = 10	0.000983	0.004117	0.021083	0.067093
N = 20	0.000754	0.003154	0.020782	0.066669
N = 50	0.000606	0.002535	0.015861	0.060334
N = 100	0.000510	0.002129	0.052025	0.232816
N = 200	0.000462	0.001931	0.058469	0.266547

The results indicate that while the training error consistently improves with a larger reservoir, the testing error does not follow the same trend. The generalization performance is strongest with a smaller reservoir of 50 neurons, after which the model begins to overfit, leading to a significant increase in testing NMSE. This analysis highlights the computational efficiency of ESNs, as high performance can be achieved without needing an excessively large and complex model.

To provide a more detailed result, we select the best case from our experiments and compare the predicted output with the golden output in both the training and testing phases. The ESN parameters for this specific case are listed in Table II.

TABLE II. Mackey-Glass ESN Model Parameters

Parameter	Value
Number of input neurons	1
Number of output neurons	1
Number of reservoir neurons	50
Spectral radius	1
Sparsity	0.6
Input scaling	8
Learning rate	0.00001

Figure 2 shows the prediction results for this configuration. The figure shows that we achieve good results in both the training and testing phases, with the model's output closely tracking the chaotic dynamics of the target signal. The ESN achieves a low NMSE of 2.22e-02 on the training data and, more importantly, maintains a similar NMSE of 2.21e-02 on the

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unseen test data, confirming its excellent prediction accuracy and ability to generalize.

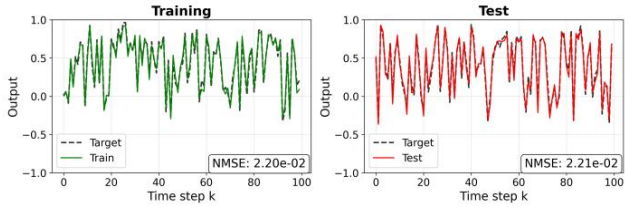


Fig. 2. ESN performance on the Mackey-Glass time-series prediction task for the best-case configuration. The left panel shows the predicted output (green) versus the target (dashed black) for the training set. The right panel shows the same for the testing set (red). The model closely follows the target signal in both phases.

B. Performance Metric

For both benchmarks, we evaluate the predictive accuracy of the ESN using NMSE. The NMSE measures the prediction error relative to the variance of the true signal, providing a scale-invariant performance metric. It is defined as:

$$\mathrm{NMSE}(y,\hat{y}) = \frac{\sum_{n=1}^{N}(\hat{y}(n)-y(n))^2}{\sum_{n=1}^{N}(y(n)-\bar{y})^2} \qquad (5)$$

where ŷ(n) is the predicted value from the ESN, y(n) is the true value from the time series, ȳ(n) is the mean of the true values, and N is the total number of time stamps in the evaluation set.

V. ESN-BASED MIMO-OFDM CHANNEL PREDICTION

In this section, we demonstrate the practical applicability of our ESN design in the context of Next-Generation (Next-G) wireless communications, focusing specifically on Multiple-Input Multiple-Output Orthogonal Frequency Division Multiplexing (MIMO-OFDM) channel prediction.

A. MIMO-OFDM Channel Prediction Task

In Next-G wireless communication, the demand for low-latency, high-reliability, and real-time connectivity continues to rise. To meet these requirements, obtaining updated and accurate channel state information (CSI) at the transmitter is essential for tasks such as resource allocation and beamforming. However, in many practical scenarios, acquiring timely CSI is challenging due to the **channel aging effect** [13]. For example, in high-mobility environments such as high-speed trains, vehicle-to-everything (V2X) communications, or low-earth-orbit (LEO) satellite links, the rapid variation of the wireless channel can cause severe delays between channel measurement and usage. Similarly, in dense urban networks with heavy user traffic and frequent handovers, CSI can quickly become outdated before it is utilized.

Channel prediction addresses the channel aging problem by forecasting the current channel state from historical, outdated CSI. Traditional approaches, such as parametric models and autoregressive (AR) models, have been widely adopted with a relatively low computational complexity. However, as wireless channels become increasingly complex with larger MIMO configurations and higher mobility, the predictive capability of these methods is significantly limited. To overcome these challenges, we propose an Echo State Network (ESN)-based method that efficiently learns the highly dynamic characteristics of wireless channels within a lightweight ML framework.

In this work, the channel data is generated using Network Simulator-3 (NS-3), which utilizes the 3GPP Spatial Channel Model (SCM) to simulate vehicle-to-vehicle (V2V) channel conditions. A MIMO system is implemented with one transmitter and one receiver, each with two antennas.

In the system, the received signals can be modeled as:

$$r(t) = H(t)x(t) + n(t) \qquad (6)$$

Where, $r(t) = [r_1(t), \dots, r_{N_r}(t)]$represents the received signal of size $N_r \times 1$ at time t, $x(t) = [x_1(t), \dots, x_{N_t}(t)]$ represents the transmitted signal of size $N_t \times 1$ at time t, and $n(t)$ represents the Gaussian noise. Channel matrix $H(t)$ denotes the channel parameters at time t with dimensions of $N_r \times N_t$.

In this way, the channel prediction can be modeled as a time-series prediction task. Assume the transmitter has access to the previous channel parameters of n time steps $\{H_1, H_2, \dots, H_n\}$. The task is to predict the coming channel H_{n+1}. And the problem can be converted into a minimization problem to minimize the difference between the actual channel and the predicted channel. Here we use NMSE $(H_{n+1}, \widehat{H}_{n+1})$ as the performance metric. The H_{n+1} represents the true value and the $\widehat{H}_{n+1}$ represents the predicted value.

B. Auto-regression for Channel Prediction

AR modeling is one of the classical approaches for channel prediction, where the impulse response of a time-varying channel is represented as a linear combination of its past states. In this framework, the current channel can be predicted by weighting and summing a sequence of previously observed channel values. To improve accuracy, Kalman filtering (KF) has been employed to estimate the AR coefficients, enabling more reliable predictions based on historical channel data.

For MIMO systems, the channels are approximated as a set of parallel SISO channels, which discards the spatial relation of the antenna arrays. A general AR-based channel predictor is as follows:

$$\widehat{H}_{n+1} = \sum_{k=0}^{p} a_k \widehat{H}_{n-k} \qquad (7)$$

Where a_k are the AR coefficients generated from historical channels and p is the order of the process.

C. ESN for Channel Prediction

In the proposed ESN method, the ESN model is used to learn the direct mapping from historical channel parameters to future channel parameters. Here, the size of the input layer and output layer of ESN is determined by the product of N_r, N_t, and the number of subcarriers. And the real and imaginary components of the complex-valued channel data are separated and stacked as the input features.

The prediction task is designed in a symbol-by-symbol manner. The channel parameters for the i-th symbol are used to

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predict the channel parameters for the (i+20)-th symbol. For our 2×2 MIMO system, the ESN was configured with a specific set of hyperparameters optimized for this application. The detailed parameters are listed in Table III. These values were selected to balance model complexity with the ability to capture the short-term temporal dependencies inherent in wireless channel fading.

TABLE III. FOR THE CHANNEL PREDICTION TASK, WIRELESS CHANNEL ESN MODEL PARAMETERS

Parameter	Value
Input neuron	64
Output neuron	64
Reservoir neuron	20
Spectral radius	1
Sparsity	0.6
Learning rate	0.00001

D. Experiment Results

To compare the performance of the proposed ESN for channel prediction tasks, an AR-based predictor has been implemented as the baseline method for comparison. Here the order of AR-based method is set to 20, allowing it to use the channels from the previous 20 time steps for prediction.

The NMSE performance between ESN-based method and AR-based method are compared with different signal-to-noise ratios (SNR), and the results are displayed in the figure.

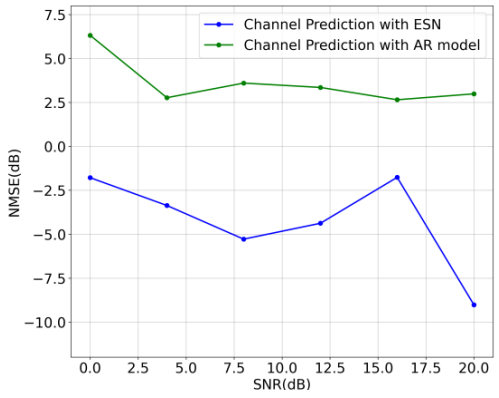


Fig. 3. NMSE performance comparison between the ESN and an AR model for wireless channel prediction across different Signal-to-Noise Ratios (SNR).

As demonstrated in the figure, the ESN-based approach consistently outperforms the traditional AR model, achieving a significantly lower NMSE across the entire range of tested SNRs. This indicates that the ESN is better equipped to learn the nonlinear patterns of the wireless channel, leading to more accurate predictions.

VI. CONCLUSION

This paper demonstrates that ESNs provide an efficient and accurate framework for time-series prediction with application in 5G/6G communications. Benchmarking on the Mackey–Glass dataset confirmed their ability to achieve low prediction error with a small number of neurons, avoiding overfitting while maintaining computational efficiency. Our simulations in MIMO-OFDM channel predictions show that ESNs consistently outperform traditional AR models across a range of SNRs. These results underscore ESN's capability to capture nonlinear channel dynamics and position it as a practical solution for intelligent, adaptive communication systems.

ACKNOWLEDGMENT

This work was supported in part by the U.S. National Science Foundation (NSF) under Grant CCF-1750450, Grant ECCS-1731928, Grant ECCS-2128594, Grant ECCS- 2314813, and Grant CCF-1937487.

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